Assignment 4: Optimizing Transformer Translation

End-to-End Hyperparameter Search for English → Hindi Translation

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github.com/ShubhamHaraniya/MLOps-Shubham-M25CSA013/tree/Assignment-4

Abstract: *A custom Encoder-Decoder Transformer (d_{model}=512) was trained on a 30% subset (3,956 pairs) of the English–Hindi parallel corpus. A fixed-hyperparameter baseline (100 epochs) achieved 44.62% BLEU and 0.0592 loss. Ray Tune (Optuna + ASHA) then searched 7 hyperparameters across 10 trials. The best configuration, trained for only 30 epochs, achieved **52.37% BLEU** (+7.75 pp over baseline) in **3.7 min** — a 5.4× speedup with 70% fewer epochs.*

1. Introduction

This assignment optimises a Transformer translation model by replacing fixed hyperparameters with automated search. Pipeline: (1) Baseline training (100 epochs, hardcoded config), (2) Ray Tune sweep (Optuna + ASHA, 7 hyperparameters, 10 trials), (3) Retrain with best config (≤30 epochs), (4) Side-by-side metric comparison.

2. Baseline Results (100 Epochs — Fixed Hyperparameters)

Trained on 3,956 sentence pairs (30% subset); English vocab: 1,712 tokens, Hindi: 1,782 tokens.

Hyperparameter	Value
Learning Rate	1×10^{-4}
Batch Size	60
Num Heads	8
d_{ff}	2048
Dropout	0.10
Num Layers	6
Epochs	100
Model Params	46,841,590

Table 1: Baseline configuration.

Metric	Value
BLEU Score	44.62%
Final Training Loss	0.0592
Training Time	19.7 min
Epochs Used	100
Dataset	30% (3,956 pairs)
Vocab EN / HI	1,712 / 1,782
Device	CUDA (P100)
Optimizer	Adam (β_1=0.9)

Table 2: Baseline evaluation metrics.

Note: Final loss 0.0592 indicates overfitting — the model memorised training pairs yet scored only 44.62% BLEU on 5 unseen validation sentences.

3. Hyperparameter Search — Ray Tune + Optuna + ASHA (10 Trials, 32.8 min)

Optuna guided Bayesian sampling; ASHA pruned poor trials after epoch 5. Best trial loss: **0.3442** at epoch 30.

Parameter	Method	Range	Rationale
`lr`	loguniform	$10^{-5} \to 10^{-3}$	Captures multiple orders of magnitude of step sizes
`batch_size`	choice	32, 64, 128	Gradient noise vs. stability
`num_heads`	choice	4, 8	Attention granularity vs. compute
`d_ff`	choice	1024, 2048, 4096	Feed-forward capacity
`dropout`	uniform	0.10 → 0.40	Regularisation; combats overfitting on small data
`num_layers`	choice	4, 6	Encoder/Decoder depth
`warmup_epo`	choice	2, 5, 10	Stable LR warm-up for higher learning rates

Table 3: Search space — 7 parameters, 10 trials, ASHA grace period = 5 epochs.

4. Best Configuration Found by Optuna

Hyperparameter	Best Value	Baseline	Reasoning
Learning Rate	**3.20e-4**	1e-4	3.2× higher; faster convergence with warmup
Batch Size	**32**	60	Smaller batch; higher gradient diversity
Num Heads	**4**	8	Sufficient for 30% dataset size
d_{ff}	**1024**	2048	Smaller FF; reduces overfitting
Dropout	**0.2543**	0.10	Higher dropout; stronger regularisation
Num Layers	**4**	6	Shallower; appropriate for dataset size
Warmup Epochs	**5**	—	Stable training with higher LR
Model Params	**23,731,958**	46,841,590	Nearly half the parameters; leaner model

Table 4: Optimal configuration from Optuna (green = tuned values). Best sweep loss: 0.3442.

5. Final Results — Baseline vs. Tuned Model

Metric	Baseline	Tuned Model	Observation
BLEU Score	44.62%	**52.37% ✓**	+7.75 pp over baseline
Final Training Loss	0.0592	0.3355	Baseline overfit; tuned is healthier
Training Time	19.7 min	**3.7 min**	5.4× speedup
Epochs Used	100	**30**	70% fewer epochs
Dataset	30%	30%	Identical — fair comparison
LR / Batch / d_{ff}	1e-4 / 60 / 2048	3.20e-4 / 32 / 1024	All optimised by Optuna
Dropout / Layers	0.10 / 6	0.2543 / 4	Higher regularisation, lighter model

Table 5: Side-by-side comparison. ✓ = target met.

The tuned model **beat the baseline BLEU (52.37% vs 44.62%)** in only 30 epochs and 3.7 min versus 19.7 min — a **5.4× speedup and 70% epoch reduction**. The higher tuned loss (0.3355 vs 0.0592) is healthy: the baseline's near-zero loss reflects overfitting, not better generalisation.

6. Conclusion

Ray Tune (Optuna + ASHA) successfully beat the baseline in far fewer resources. Key findings: (1) ASHA pruning focused compute on promising trials; (2) Optuna selected a leaner, better-regularised model (23.7M vs 46.8M params); (3) Baseline's 0.0592 loss was misleading overfitting — the tuned model generalises better; (4) Automated search over 7 hyperparameters in 32.8 min outperformed 100 epochs of manual tuning.

References

[1] A. Vaswani et al., "Attention Is All You Need," *NeurIPS*, 2017.
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[3] T. Akiba et al., "Optuna: A Next-generation Hyperparameter Optimization Framework," *Proc. ACM KDD*, 2019.